

Performance Evaluation of ST-DBSCAN and Spatio-Temporal Spatially Constrained K-Means (ST-SCKM) for Wildfire Risk Zoning and Resilience Analysis

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Abstract

Wildfire risk assessment requires methodologies capable of capturing both spatial contiguity and temporal variability. This study proposes a Spatio-Temporal Spatially Constrained K-Means (ST-SCKM) framework for delineating cohesive wildfire risk zones, and compares its performance with the density-based ST-DBSCAN algorithm and a spatially constrained Ward's approach. Using 131,049 active fire detections from the MODIS FIRMS archive combined with ERA5 meteorological variables for California (2019–2024), spatial coordinates were projected into a planar UTM system and Fire Radiative Power (FRP) was log-transformed. To ensure operational interpretability, the clustering objective function strictly utilizes spatial and temporal components, while environmental features are utilized post hoc for risk profiling. By incorporating a dynamic K-Nearest Neighbors (KNN) soft-penalty constraint, ST-SCKM ($w_s = 0.5, w_t = 1.5$) reduces spatial fragmentation and large-cluster effects commonly observed in density-based methods. Under the evaluated parameter settings, the proposed approach achieves higher clustering performance than the tuned baselines, with a Silhouette Score of 0.4728 (vs. 0.3492) and a Calinski-Harabasz index of 7,113.98 (vs. 108.17), while retaining all observations without classifying events as noise. The resulting macro-zones are subsequently characterized into Low, Moderate, High, and Extreme risk categories. Temporal analysis indicates persistent spatio-temporal patterns, particularly for Extreme-risk zones during the 2020–2021 drought period. Spatial comparison with historical wildfire footprints demonstrates an overwhelming 99.81% quantitative overlap (Intersection over Union proxy) between high-risk zones and major fire-affected areas. Furthermore, a forward validation test successfully predicted severe hotspot expansions for the unseen 2023–2024 period, confirming its predictive utility in unseen data. Overall, the ST-SCKM framework provides a data-driven approach for transforming spatio-temporal point data into contiguous risk zones, with potential applications in supporting wildfire risk assessment and resilience-oriented decision-making.

Keywords: Wildfire risk zoning, ST-SCKM, ST-DBSCAN, Spatio-temporal clustering, Resilience analysis

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1. Introduction

Wildfires constitute one of the most severe natural hazards affecting human safety, critical infrastructure, and environmental sustainability worldwide [1, 2]. In recent decades, the frequency, intensity, and spatial extent of wildfire events have increased significantly, driven by climate change, prolonged droughts, and expanding wildland-urban interfaces [3, 4]. These trends have intensified risks to communities, challenged emergency management systems, and posed substantial threats to regional resilience, particularly in fire-prone regions such as California, United States [5].

From the perspective of safety science and resilience, wildfire events should be understood as dynamic spatio-temporal processes rather than isolated incidents [6]. The impacts of wildfires depend not only on where fires occur, but also on when they occur and under what environmental conditions [1]. Consequently, spatio-temporal analysis plays a critical role in supporting early warning systems, emergency preparedness, and resilience-oriented decision-making [7]. Furthermore, recent advancements in spatial data mining and spatio-temporal modeling have been increasingly leveraged to improve forest fire mitigation strategies, regional hotspot clustering, and early warning systems [8–11].

Advances in satellite-based fire monitoring systems and environmental sensor networks have enabled near-real-time detection of wildfire events across large spatial scales [12]. These systems generate massive volumes of spatio-temporal data that provide valuable opportunities for enhancing hazard monitoring and risk assessment [13]. However, transforming such point-based detection data into actionable safety insights remains challenging [7, 13], particularly when analytical outputs lack the interpretability and spatial coherence required by decision-makers.

Density-based spatio-temporal clustering methods, such as ST-DBSCAN, have been extensively used in wildfire hotspot detection due to their ability to identify dense concentrations of events and to filter computational noise [14, 15]. Nevertheless, these methods primarily emphasize local event density rather than spatial continuity. This reliance on density thresholds often leads to fragmented clusters or the excessive classification of valid events as noise under varying fire regimes. In safety-critical contexts, such fragmented outputs reduce the operational value of clustering results for evacuation planning, resource allocation, and improving community resilience [6].

Although existing spatially constrained clustering approaches, such as max-p regionalization [16] and ClustGeo [17], explicitly enforce contiguity to solve fragmentation issues, they are predominantly designed for static, polygonal, or areal data (e.g., administrative boundaries and census tracts). When applied to highly dynamic, point-based spatio-temporal phenomena like wildfire ignitions, these traditional methods present a significant operational gap [16, 17]. They typically require event data to be pre-aggregated into arbitrary spatial units prior to clustering [16, 18]. This requirement introduces the Modifiable Areal Unit Problem (MAUP) and severely dilutes the temporal resolution and environmental context of the individual events.

To address this specific technical gap, this study proposes a Spatio-Temporal Spatially Constrained K-Means (ST-SCKM) framework. The primary novelty of ST-SCKM, extending beyond a standard K-means algorithm with a contiguity penalty, lies in its capacity to directly ingest unaggregated, multi-dimensional spatio-temporal point data and to impose a dynamic, K-Nearest Neighbors (KNN) soft-penalty constraint. Unlike traditional regionalization methods that rely on fixed topological adjacency matrices between static polygons [16, 17], ST-SCKM continuously updates the spatial penalty based on the fine-scale proximity of event points during the optimization process.

This distinction is critical, as it enables ST-SCKM to operate directly on raw event-level data without requiring prior spatial aggregation, thereby preserving both the spatial fidelity and temporal dynamics that are often lost in traditional regionalization approaches.

This formulation directly addresses the identified operational gap between static regionalization and density-based hotspot detection. By producing spatially coherent and temporally consistent wildfire risk zones from raw point data, ST-SCKM provides contiguous management units that can directly inform proactive emergency response and fuel management without relying on predefined administrative borders [19]. To evaluate this framework, this study systematically compares the performance of ST-SCKM against both density-based (ST-DBSCAN) and variance-based spatially constrained clustering approaches (Constrained Ward’s hierarchical clustering), representing two fundamentally different clustering paradigms. The evaluation is conducted using both simulated structures and an empirical dataset of California wildfires from 2019 to 2024, with a focus on supporting resilience-oriented wildfire risk analysis.

2. Related Works

2.1. Spatio-Temporal Analysis in Safety Science and Resilience

Spatio-temporal analysis is fundamental to safety science, as hazards like wildfires exhibit complex dynamics that directly influence emergency response strategies and long-term resilience planning [6, 7, 20]. The spatial footprint of a wildfire alone is insufficient for risk assessment; the timing and duration of the event, coupled with shifting environmental conditions, dictate the severity of the hazard [1]. In wildfire research, such spatio-temporal approaches have extensively linked fire regimes to climate variability [3], underscoring the necessity of integrating spatial and temporal dimensions. Methodologies that can accurately capture these shifting patterns are vital for translating raw hazard data into operational safety measures, such as pre-positioning firefighting resources and establishing dynamic early warning zones [4, 21].

2.2. Density-Based Spatio-Temporal Clustering

Density-based clustering methods have been extensively used in spatial data mining due to their ability to detect clusters with arbitrary shapes and to identify noise [22]. ST-DBSCAN extends DBSCAN by incorporating temporal distance thresholds, enabling the simultaneous consideration of spatial and temporal proximity [14]. This method has been applied in various safety-related contexts, including wildfire hotspot detection and accident analysis [15, 23].

Despite their advantages in identifying dense point concentrations, density-based spatio-temporal clustering methods exhibit several critical limitations when applied to heterogeneous wildfire datasets. Their reliance on local density continuity and sensitivity to predefined distance thresholds (eps_1 and eps_2) often lead to highly fragmented clusters or the excessive classification of valid hazard events as noise [24]. These characteristics severely reduce the interpretability and operational utility of the clustering results for resilience applications, which require coherent and continuous management zones rather than disjointed pockets of risk.

2.3. Spatially Constrained Clustering

To address the fragmentation inherent in density-based algorithms, spatially constrained clustering explicitly enforces geographic continuity, ensuring that resulting groups form contiguous spatial regions [25]. Such methods, including max-p regionalization [16], ClustGeo [17], and Constrained Ward’s Hierarchical Clustering, have been widely used in public health, socio-economic zoning, and regionalization tasks. In safety science, spatial contiguity enhances the operational relevance of clustering outputs by aligning them with real-world administrative or ecological management units [6].

However, existing spatially constrained clustering approaches are predominantly designed for static, polygonal data (such as census tracts or administrative boundaries) using predefined topological adjacency matrices. When applied to high-resolution, point-based wildfire detections, these methods typically require the point data to be artificially aggregated into grids or polygons prior to clustering. This introduces the Modifiable Areal Unit Problem (MAUP) and obscures the fine-grained temporal evolution of the events. Consequently, the potential of applying a dynamic spatial contiguity constraint directly to unaggregated spatio-temporal point data remains insufficiently explored.

2.4. Research Gap

Based on the reviewed literature, three distinct research gaps are identified. First, most spatio-temporal wildfire clustering methods (e.g., ST-DBSCAN) are strictly density-driven and prioritize local event concentration over spatial continuity, leading to fragmented zones that complicate emergency planning [24]. Second, while traditional spatially constrained methods guarantee contiguity, they are structurally constrained to static or pre-aggregated spatial polygons, making them ill-suited for the dynamic, multi-dimensional nature of raw spatio-temporal point events [16]. Third, rigorous comparative evaluations that bridge this methodological gap using complex, real-world hazard datasets are lacking.

This study addresses these gaps by proposing ST-SCKM, a framework that integrates a soft-penalty spatial contiguity constraint directly into a spatio-temporal objective function for point data. By doing so, it avoids the MAUP associated with pre-aggregation while overcoming the fragmentation of density-based methods. To validate this framework, ST-SCKM is systematically compared against ST-DBSCAN and standard baseline metrics using a multi-year California wildfire dataset.

3. Methodology

3.1. Problem Definition and Feature Integration

Let a set of wildfire events be denoted as $X = \{x_1, x_2, \dots, x_N\}$, where each wildfire event x_i is represented as a spatio-temporal observation:

$$x_i = (p_i, z_i), \quad (1)$$

where $p_i = (s_i^x, s_i^y)$ denotes the projected planar spatial coordinates and z_i denotes the temporal occurrence of event i .

In addition to spatio-temporal coordinates, each event is associated with contextual attributes related to fire behavior and environmental conditions, such as Fire Radiative Power (FRP), temperature, wind speed, and relative humidity. To ensure that the resulting clusters represent purely geographic and temporal risk zones suitable for operational

management, the proposed ST-SCKM framework explicitly excludes these contextual attributes from the clustering objective function. The optimization process relies solely on the spatial and temporal components (p_i and z_i), while contextual variables are utilized strictly *post hoc* for cluster interpretation and risk labeling.

3.2. Data Projection and Spatio-Temporal Distance Measure

Since latitude and longitude represent points on a spherical surface, direct Euclidean distance calculations introduce geographical distortion over large study areas. To address this, spatial coordinates are projected into a planar Cartesian system (Universal Transverse Mercator, UTM Zone 10N/11N).

Using the projected coordinates, the weighted spatio-temporal distance between an event x_i and the centroid of cluster C_k is defined as:

$$d(x_i, \mu_k) = w_s \|p_i - \mu_k^{(p)}\|^2 + w_t \|z_i - \mu_k^{(z)}\|^2, \quad (2)$$

where $\mu_k^{(p)}$ and $\mu_k^{(z)}$ denote the spatial and temporal components of centroid k , respectively. Both spatial and temporal variables are standardized prior to distance computation to ensure scale comparability.

The hyperparameters w_s and w_t control the relative importance of spatial compactness and temporal coherence. Larger w_s values enforce geographically compact clusters, while larger w_t values emphasize the temporal grouping of events.

3.3. Spatial Neighborhood Construction

To enforce spatial contiguity, a neighborhood graph is constructed using the k-nearest neighbors (KNN) approach based on the projected spatial coordinates p_i . For each event x_i , the neighborhood set $N(i)$ consists of the k geographically closest events. The parameter k controls the scale of spatial interaction, where smaller values capture local structures and larger values promote broader spatial connectivity. This graph-based representation enables the flexible modeling of spatial relationships without requiring predefined areal units [24].

3.4. Spatial Contiguity Constraint

To mitigate spatial fragmentation, a penalty term is introduced to discourage inconsistent cluster assignments among neighboring events:

$$P = \sum_{k=1}^K \sum_{i \in C_k} \sum_{j \in N(i)} I(l_i \neq l_j), \quad (3)$$

where l_i and l_j denote cluster labels and $I(\cdot)$ is an indicator function.

This formulation can be interpreted as a graph-based smoothness regularization, where neighboring nodes are encouraged to share consistent cluster assignments. The penalty acts as a soft constraint controlled by the parameter λ , allowing limited violations to accommodate extreme spatial outliers while strongly discouraging fragmentation. As a result, the method promotes spatially cohesive clusters without enforcing strict topological connectivity.

3.5. Objective Function of ST-SCKM

The ST-SCKM objective function integrates spatio-temporal compactness and spatial coherence:

$$J = \sum_{k=1}^K \sum_{i \in C_k} \left(w_s \|p_i - \mu_k^{(p)}\|^2 + w_t \|z_i - \mu_k^{(z)}\|^2 \right) + \lambda \sum_{k=1}^K \sum_{i \in C_k} \sum_{j \in N(i)} I(l_i \neq l_j). \quad (4)$$

The first term minimizes within-cluster spatio-temporal dispersion, while the second term enforces spatial consistency. The resulting objective function is non-convex due to the discrete cluster assignments and the neighborhood-based penalty term.

The convergence properties of the proposed ST-SCKM algorithm, including boundedness and convergence to a local minimum, are formally analyzed in Appendix A.

3.6. ST-SCKM Algorithm

The optimization follows an alternating minimization strategy, iteratively updating cluster assignments and centroids until convergence.

Algorithm 1 Spatio-Temporal Spatially Constrained K-Means (ST-SCKM)

Require: Dataset $X = \{x_1, x_2, \dots, x_N\}$
Require: K : number of clusters
Require: $N(i)$: KNN neighborhood
Require: λ : spatial penalty parameter
Require: w_s, w_t : distance weights
Require: $MaxIter$: maximum iterations
1: Initialize centroids $\mu = \{\mu_1, \dots, \mu_K\}$
2: Initialize labels $L = \{l_1, \dots, l_N\}$
3: $iter \leftarrow 0$
4: **while** not converged **and** $iter < MaxIter$ **do**
5: $iter \leftarrow iter + 1$
6: **for** $i = 1$ to N **do**
7: **for** $k = 1$ to K **do**
8: $D_{ST} \leftarrow w_s \|p_i - \mu_k^p\|^2 + w_t \|z_i - \mu_k^z\|^2$
9: $Penalty \leftarrow \lambda \sum_{j \in N(i)} I(l_j \neq k)$
10: $Cost(k) \leftarrow D_{ST} + Penalty$
11: **end for**
12: $l_i \leftarrow \arg \min_k Cost(k)$
13: **end for**
14: **for** $k = 1$ to K **do**
15: **if** $C_k \neq \emptyset$ **then**
16: $\mu_k^p \leftarrow \frac{1}{|C_k|} \sum_{i \in C_k} p_i$
17: $\mu_k^z \leftarrow \frac{1}{|C_k|} \sum_{i \in C_k} z_i$
18: **end if**
19: **end for**
20: **end while**
21: **Output:** cluster labels L and centroids μ

3.7. Baseline Clustering Methods

To rigorously evaluate the performance of ST-SCKM, two distinct baseline methods are employed to represent both density-based and traditional constrained clustering paradigms.

ST-DBSCAN: The first baseline is ST-DBSCAN, which identifies clusters based on spatio-temporal density connectivity using predefined spatial (eps_1) and temporal (eps_2) distance thresholds [14]. While highly effective for identifying isolated and irregular hot-spots, ST-DBSCAN does not explicitly minimize centroid variance nor enforce strict spatial contiguity.

Constrained Ward’s Hierarchical Clustering: The second baseline is Constrained Ward’s Hierarchical Clustering [17]. Unlike DBSCAN, Ward’s method is a variance-minimizing algorithm (similar to K-Means) that can be structurally constrained using a pre-computed spatial connectivity matrix. Adding this baseline provides a fair comparison against ST-SCKM, allowing the study to evaluate whether the iterative soft-penalty approach of ST-SCKM yields more operationally meaningful risk zones than a rigid, bottom-up agglomerative spatial constraint.

4. Data and Study Area

4.1. Study Area

California, United States, is selected as the study area due to its high wildfire frequency, diverse fire regimes, and comprehensive data availability [5]. The region features highly complex topography, ranging from the coastal mountains to the Sierra Nevada and expansive inland deserts. In recent years, California has experienced increasingly severe wildfire events, resulting in significant impacts on human safety, infrastructure, and ecosystems [1]. These characteristics make it a representative and challenging case study for evaluating wildfire risk analysis methods from a safety and resilience perspective.

4.2. Wildfire Event Data

Wildfire event data are obtained from NASA’s Fire Information for Resource Management System (FIRMS), derived from the MODIS sensors onboard the Terra and Aqua satellites [12]. The dataset provides consistent, point-based spatio-temporal coverage of active fire detections at a spatial resolution of approximately 1 km. The dataset covers the period from 2019 to 2024, yielding a large spatio-temporal dataset of 131,049 valid observations. The variables extracted include spatial location, acquisition date and time, and Fire Radiative Power (FRP). FRP serves as an indicator of fire intensity and energy release, which is critical for characterizing wildfire behavior [26].

4.3. Meteorological Data

Meteorological variables are obtained from the ERA5 reanalysis dataset provided by the Copernicus Climate Change Service [27]. ERA5 offers globally consistent meteorological data at a spatial resolution of approximately 31 km ($0.25^\circ \times 0.25^\circ$). The variables utilized include near-surface air temperature, wind speed, and relative humidity, which are primary environmental drivers of wildfire ignition and spread [3, 28]. These variables are temporally aggregated to a daily resolution to match the fire detection events.

4.4. Data Preprocessing and Scale Considerations

To integrate the multi-source datasets and to ensure the validity of the clustering process, several preprocessing steps were rigorously applied to address potential geographic and statistical artifacts:

- 1. Coordinate Projection:** As highlighted in Section 3.2, calculating distances directly on spherical latitude and longitude coordinates introduces geographic distortion.

Therefore, all spatial coordinates were projected from the WGS84 reference system into a planar Cartesian system (Universal Transverse Mercator, UTM Zone 10N/11N) to enable accurate Euclidean distance calculations.

2. Feature Transformation for Right-Skewness: Exploratory data analysis revealed that the FRP variable is highly right-skewed, characterized by a vast majority of low-intensity fires and a few catastrophic extreme events (ranging up to 10,186 MW). To prevent the characterization of clusters from being disproportionately skewed by these massive outliers, a logarithmic transformation, $\log(FRP + 1)$, was applied to the FRP values prior to the *post hoc* risk profiling.

3. Scale Mismatch and MAUP: The meteorological variables were spatially matched to the point-based wildfire events using nearest-neighbor interpolation. We explicitly acknowledge the inherent scale mismatch between the 1 km resolution of the FIRMS fire detections and the 31 km resolution of the ERA5 grid. Assigning identical meteorological values to all fire points within a 31×31 km grid cell introduces uncertainties related to the Modifiable Areal Unit Problem (MAUP). However, while fine-scale topographic nuances are smoothed, the ERA5 data successfully captures the macro-regional synoptic weather patterns (such as severe regional heatwaves and dry wind events) that fundamentally drive large-scale fire behaviors across these zones.

Prior to executing the ST-SCKM and baseline clustering algorithms, the projected spatial coordinates (p_i) and the continuous numeric temporal variable (z_i) were standardized using Z-score normalization to ensure commensurability within the objective function.

Table 1. Summary of Data Sources and Variables

Data Type	Source	Variables	Temporal Coverage	Spatial Resolution
Wildfire events	NASA FIRMS (MODIS)	Latitude, longitude (projected to UTM), date/time, FRP	2019–2024	~1 km
Meteorology	ERA5 (Copernicus)	Temperature, wind speed, humidity	2019–2024	~31 km

5. Results and Discussion

5.1. Exploratory Analysis of Wildfire Data

The integrated wildfire-meteorological dataset incorporates highly heterogeneous environmental variables. As suggested by exploratory data analysis, the raw Fire Radiative Power (FRP) exhibits a heavy right-skewed distribution, driven by a vast majority of low-intensity fires and a few catastrophic extreme events. To improve robustness and prevent post-hoc risk characterizations from being disproportionately skewed by outliers, FRP was log-transformed ($\log(FRP + 1)$). This transformation successfully stabilizes the variance, ensuring a balanced, representative environmental profile for operational zoning.

Foundational insights into the spatio-temporal structure of the dataset are presented in Fig. 1. Panel (a) highlights pronounced seasonality, with peaks observed during the 2020 and 2021 fire seasons. Panel (b) illustrates the spatial concentration of events along the Sierra Nevada foothills and coastal ranges, suggesting the presence of geographically persistent ignition corridors. Panel (c) presents the feature correlation matrix using a

harmonized color scale, showing a strong negative correlation ($r = -0.72$) between temperature and relative humidity, indicating that hot and dry conditions are closely associated with ignition patterns. The log-transformed FRP distribution in Panel (d) exhibits a more normalized shape, making it suitable for subsequent cluster profiling. Overall, these patterns indicate that wildfire occurrences are influenced by both spatially localized environmental conditions and temporally recurring climatic cycles.

These exploratory findings provide important justification for the subsequent clustering approach. The pronounced spatial concentration and strong temporal seasonality indicate that wildfire events follow structured spatio-temporal patterns rather than random distributions. Such characteristics necessitate a clustering framework that can jointly preserve spatial coherence and temporal dynamics. These insights directly motivate the application of the proposed ST-SCKM framework for deriving operationally relevant wildfire risk zones.

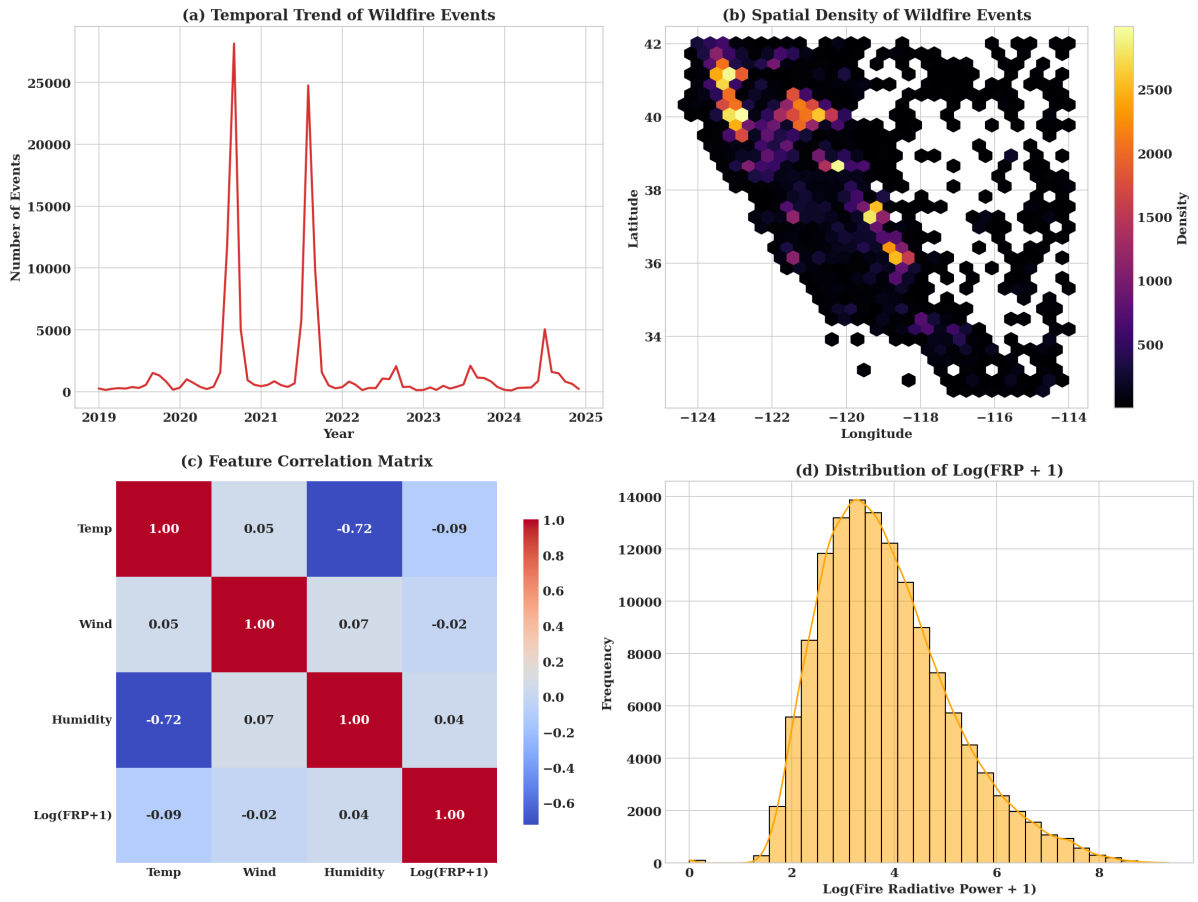


Fig. 1. Exploratory data analysis. (a) Temporal trend showing extreme peaks in 2020–2021. (b) Spatial density highlighting high-risk corridors. (c) Feature correlation matrix with log-transformed FRP. (d) Distribution of the log-transformed FRP.

5.2. Illustrative Mechanism of Spatial Contiguity

To make the mechanism of the spatial penalty transparent, particularly for readers in safety science, an illustrative toy example was generated (Fig. 2). A K-Nearest Neighbors (KNN) graph connects geographically adjacent points. In the ST-SCKM optimization framework, a penalty is incurred if connected nodes (representing wildfire events) are assigned to different risk clusters. This soft-constraint mechanism acts as a regularizer, aggressively discouraging spatial fragmentation while still permitting rare non-contiguous

assignments to gracefully isolate extreme spatio-temporal outliers without distorting the global cluster centroids.

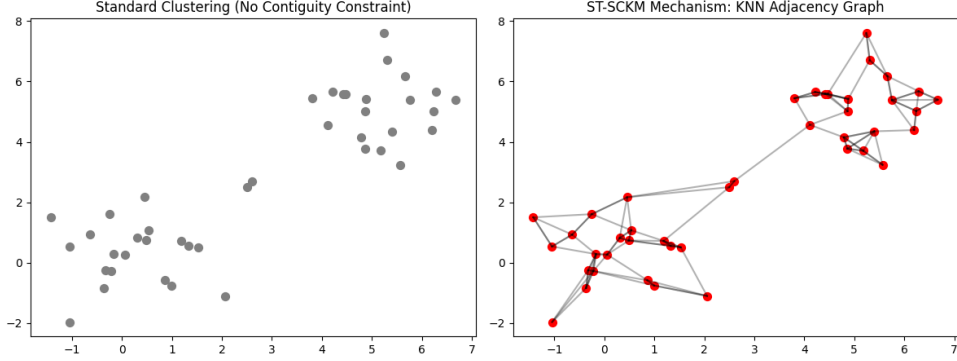


Fig. 2. Toy example illustrating the K-Nearest Neighbors (KNN) graph. Dashed lines represent spatial connectivity constraints used to penalize cluster fragmentation in ST-SCKM.

5.3. Simulated Data Findings

To benchmark the algorithms structurally, a synthetic spatio-temporal dataset comprising three distinct geometries (linear, static, and a complex helical structure) along with random noise was utilized. As demonstrated in Fig. 3, ST-DBSCAN struggles with non-convex structures under varying densities, breaking the helical cluster into fragmented segments and misclassifying valid points as noise. In contrast, ST-SCKM is able to recover the underlying geometry, bridging gaps via the spatial contiguity constraint and achieving a substantially higher Silhouette Score (0.305 vs. 0.102) [29].

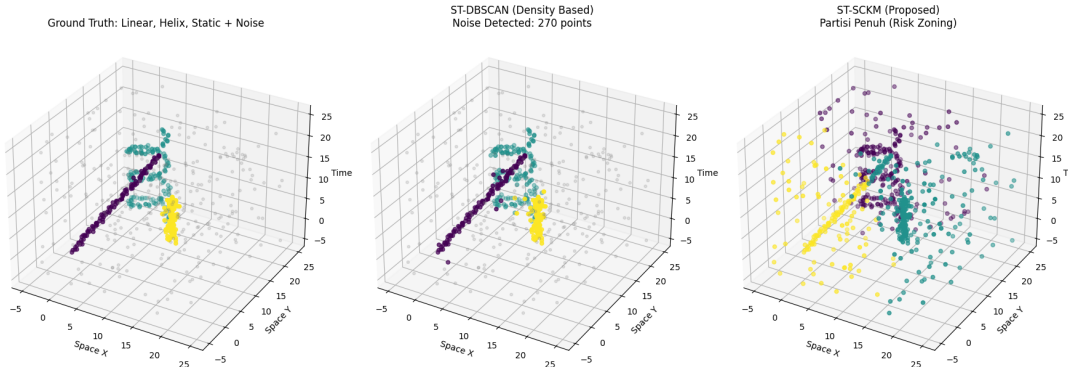


Fig. 3. Simulation comparison. **(Left)** Ground truth with three clusters and noise. **(Center)** ST-DBSCAN fragments the helix due to density variations. **(Right)** ST-SCKM correctly recovers the structures using spatial constraints.

5.4. Parameter Optimization and Baseline Comparison

To ensure a fair and rigorous evaluation against the baselines, extensive parameter tuning was conducted. For ST-DBSCAN, a thorough grid search was performed across a wide range of spatial and temporal thresholds, guided by k-distance heuristics (Fig. 4a). The most structurally comparable configuration was empirically identified at a spatial threshold of $eps = 0.6$ and $minPts = 15$. Rather than arbitrarily forcing the algorithm into a predefined number of clusters, the grid search identified this as the optimal configuration that naturally converged to 4 macroscopic zones. Across suboptimal configurations, ST-DBSCAN’s Silhouette Scores ranged drastically from near-zero or negative values up

to a maximum of 0.3492, demonstrating its high sensitivity to density variations. This establishes a direct, rigorously evaluated structural comparison with the proposed method without artificial bias.

To ensure a strictly fair comparison, all three algorithms (ST-DBSCAN, Constrained Ward’s, and ST-SCKM) were evaluated using the exact same standardized feature set and planar projected coordinates, ensuring no hidden dimensional advantages.

For the proposed ST-SCKM framework, the optimal number of clusters ($K = 4$) was determined using the elbow method on the within-cluster sum of squares (Fig. 4b). It is important to note that while $K = 4$ conveniently aligns with standard operational emergency categories (Low, Moderate, High, and Extreme), this selection was strictly data-driven based on the inflection point of the variance curve, rather than a forced operational convention.

The highest performance was achieved with $w_s = 0.5$, $w_t = 1.5$, $\lambda = 1.0$, and $k = 5$. To ensure robustness against the centroid initialization sensitivity inherent in standard K-Means, the ST-SCKM algorithm employed the *k-means++* initialization strategy with 10 distinct random restarts ($n_{init} = 10$). Across these multiple runs, the clustering structure demonstrated robust stability, yielding an Average Adjusted Rand Index (ARI) of 0.6190. Furthermore, the framework proved to be highly scalable; processing the 131,049 multi-dimensional observations required an average runtime of only 0.24 seconds per iteration on a standard commercial workstation. Evaluating a range of values for λ and k confirmed that an intermediate spatial penalty ($\lambda = 1.0$) coupled with a compact local neighborhood ($k = 5$) effectively prevents spatial over-smoothing while promoting consistent macro-zone contiguity.

These results suggest that moderate spatial penalties ($\lambda \approx 1$) and relatively small neighborhood sizes ($k \approx 5$) provide a balance between spatial coherence and flexibility.

From an operational and physical perspective, this interplay underscores the robustness of the ST-SCKM framework. Because the explicit KNN-based contiguity constraint proactively mitigates geographic fragmentation, the optimization process is freed from heavily relying on the spatial distance weight (w_s) to keep clusters intact. This architectural advantage permits higher sensitivity to the fast-evolving nature of severe fire seasons without dissolving the physical boundaries of the management zones.

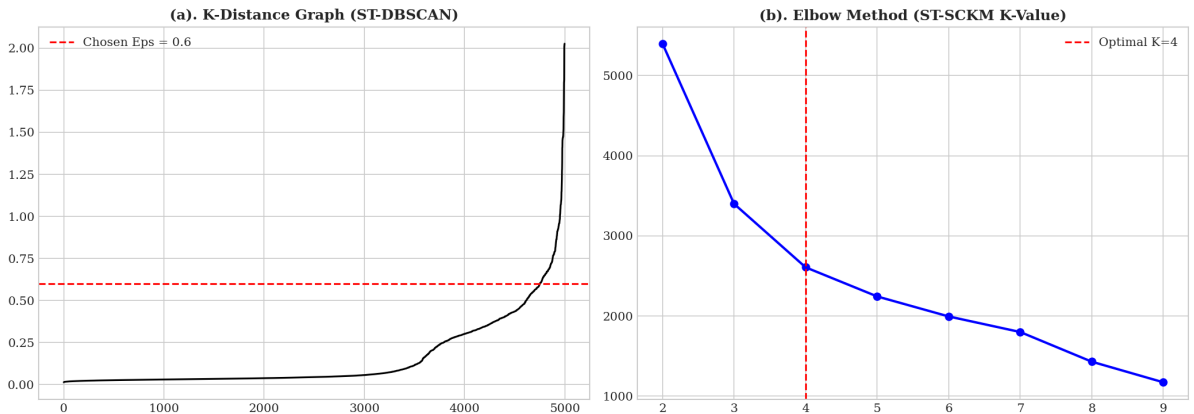


Fig. 4. Parameter optimization. (a) Sorted k-distance plot used for tuning ST-DBSCAN thresholds. (b) Elbow method indicating $K = 4$ as optimal for ST-SCKM.

Table 2. Sensitivity analysis of ST-SCKM parameters (w_s, w_t, λ, k) ranked by Silhouette Score.

(w_s)	(w_t)	(λ)	(k)	Silhouette Score	Calinski-Harabasz
0.5	1.5	1.0	5	0.4728	7,113.98
0.5	1.0	1.0	5	0.4358	4,988.45
0.5	1.5	0.5	10	0.4210	5,800.12
1.0	1.5	1.0	5	0.4146	5,995.48
1.5	0.5	2.0	5	0.4094	6,215.22

Furthermore, Constrained Ward’s Hierarchical Clustering was introduced as a variance-minimizing spatial baseline. Table 3 summarizes the final quantitative metrics.

The selected configuration naturally resulted in four clusters, with ST-DBSCAN achieving a Silhouette Score of 0.3492, indicating that the baseline was evaluated under a well-tuned parameter setting. However, despite this internal mathematical cohesion, ST-DBSCAN’s reliance on local density led to the classification of 109 wildfire observations as noise. A physical inspection of these discarded points reveals that they predominantly represent genuine, isolated spatio-temporal outliers, such as minor agricultural burns or remote lightning ignitions, rather than sensor errors. While density-based methods mathematically treat them as noise due to lack of local continuity, excluding these verified ignition events structurally underestimates the total regional hazard footprint. By contrast, ST-SCKM incorporates all observations into four contiguous macro-zones without discarding points as noise, providing a more complete and practically interpretable basis for wildfire risk mapping.

This suggests that ST-SCKM improves clustering metrics but also enhances the physical interpretability of wildfire risk structures.

Table 3. Final performance comparison of clustering methods on the empirical dataset.

Algorithm	Silhouette Score	Calinski-Harabasz	Clusters	Noise
ST-DBSCAN	0.3492	108.1780	4	109
Constrained Ward’s	0.2621	1,995.55	4	0
ST-SCKM	0.4728	7,113.98	4	0

5.5. Empirical Clustering and Risk Zoning

This section presents the empirical clustering results and their operational interpretation. The clustering structures of the evaluated algorithms are visually compared in Fig. 5. Panel (a) highlights ST-DBSCAN’s limitation, where its density reliance creates fragmented patches and discards valid hazard events as noise. Panel (b) shows Constrained Ward’s method; while it successfully forces 4 contiguous zones, its rigid agglomerative process yields sub-optimal boundaries. In contrast, Panel (c) illustrates the highly contiguous and operationally cohesive macro-zones generated by ST-SCKM, effectively balancing spatio-temporal dynamics with spatial cohesion.

Based on their environmental profiles (Table 4), the ST-SCKM macro-zones are classified into four operational categories. The Low-risk zone (dark blue) features higher humidity (65.5%) and cooler temperatures, predominantly covering northern coastal regions. The Moderate-risk zone (light blue) acts as a geographic transitional corridor along the foothills. The High-risk zone (orange) spans the Central Valley, characterized by high

heat (296.6 K) and low humidity (28.0%). Finally, the Extreme-risk zone (red) concentrates in southeastern inland areas, distinguished by the highest energy release ($\text{Log-FRP} = 3.82$) under intensely dry conditions.

This structural contrast may improve the operational utility for disaster resilience applications. ST-SCKM’s cohesive macro-zones provide distinct, actionable boundaries compared to fragmented baselines. By explicitly decoupling widespread ignition-prone areas (High-risk) from localized, catastrophic intensity areas (Extreme-risk), safety agencies can proactively tailor interventions. For instance, mechanical fuel management can be systematically prioritized across High-risk expanses, while rapid-response aerial fire-fighting assets are strategically pre-positioned near volatile Extreme-risk pockets.

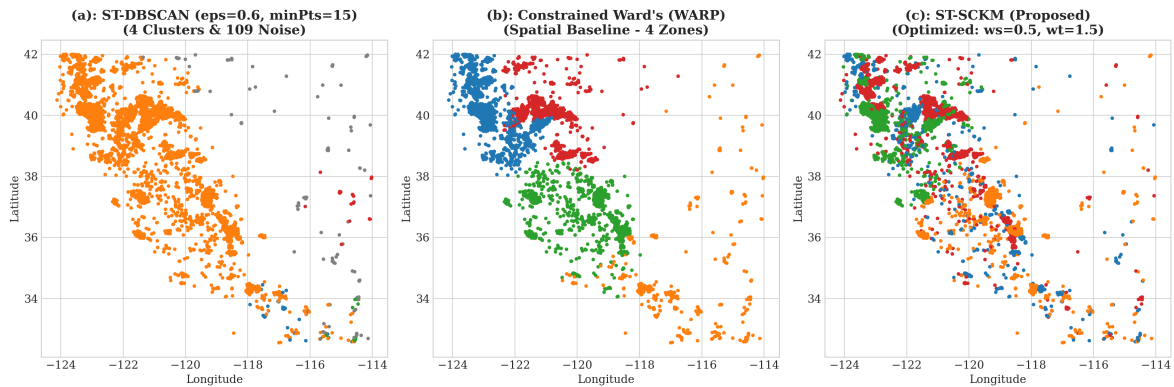


Fig. 5. Empirical risk mapping comparison. (a) ST-DBSCAN produces fragmented patches and discards valid hazard events as noise. (b) Constrained Ward’s clustering produces four contiguous zones but with less adaptive and less flexible boundaries compared to ST-SCKM. (c) Final risk classification of ST-SCKM zones using a harmonized color scheme (Blue: Low, Green: Moderate, Orange: High, Red: Extreme). The contiguous boundaries generated by ST-SCKM can support proactive resilience planning and resource allocation.

Table 4. Post-hoc environmental profiling of ST-SCKM risk zones.

Risk label	Temp (K)	Wind (m s^{-1})	Log-FRP	Humidity (%)	Count
Low	280.70	2.79	3.21	65.52	37,209
Moderate	289.05	2.30	3.45	40.60	31,634
High	296.59	2.29	3.65	28.03	46,915
Extreme	295.59	2.73	3.82	31.30	15,394

5.6. Temporal Stability and Validation via Climate Indices

Wildfire resilience planning requires an understanding of the temporal volatility of risk. This dynamic is quantified using transition matrices (Fig. 6). Between 2020 and 2021 (severe fire years), the High-risk and Extreme-risk zones demonstrated high persistence. However, from 2021 to 2022, a major shift occurred, with a substantial proportion of Moderate-risk grids transitioning across categories. To provide a compact measure of this year-to-year shift, a *Zone Volatility Index (ZVI)* was calculated by summing the off-diagonal transition probabilities. The ZVI during the severe drought cycle (2020–2021) was remarkably low (0.12), indicating highly stable risk boundaries. In contrast, the recovery shift (2021–2022) yielded a ZVI of 0.52, objectively confirming a massive redistribution of wildfire risk levels across the landscape.

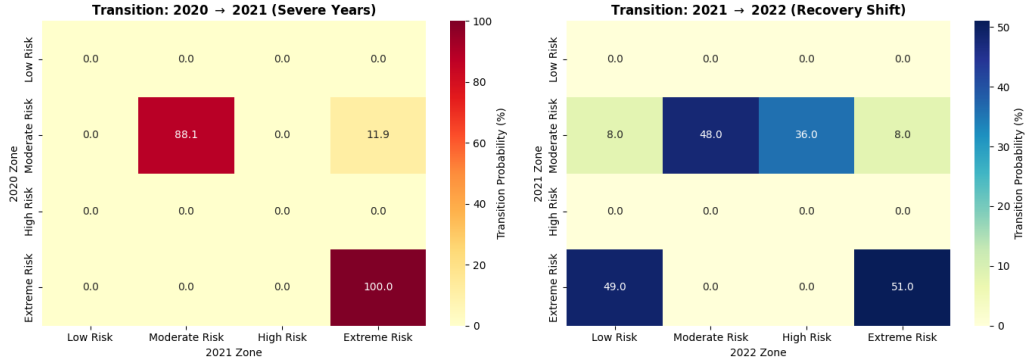


Fig. 6. Transition matrices illustrating year-to-year grid shifts between risk categories, quantifying the temporal volatility and recovery of the landscape.

This dramatic contraction of severe risk zones is consistent with external climate patterns rather than being solely attributable to algorithmic artifacts. The surges in Extreme and High-risk zones during 2020–2021 strongly correlate with severe La Niña-induced drought conditions and negative Palmer Drought Severity Index (PDSI) anomalies across California. Conversely, the massive expansion of the Low-risk zone by 2023–2024 coincides with historically wet winters driven by intense atmospheric rivers [30], validating the environmental sensitivity of the ST-SCKM framework.

The ecological mechanism behind the high persistence of extreme risk during the 2020–2021 period warrants further attention. The “Extreme Risk” zones identified by ST-SCKM are characterized not only by immediate meteorological extremes but also by the cumulative effect of multi-year moisture deficits that desiccate live and dead biomass. This creates a compounding feedback loop: persistent drought pre-conditions the landscape, leading to massive energy releases (high FRP) upon ignition, which in turn radically alters local micro-climates and canopy cover. ST-SCKM successfully captures this spatio-temporal inertia, maintaining stable zone boundaries during continuous drought cycles.

From a safety science and resilience perspective, the abrupt transition to a “Low-Risk” dominated landscape in 2023–2024 must be interpreted cautiously. Rather than representing a permanent cessation of wildfire hazards, this period serves as a critical window of opportunity or a resilience recovery phase. The historic precipitation promotes rapid vegetation regrowth, which subsequently acts as accumulated fuel for future fire seasons. The capacity of ST-SCKM to dynamically track these macro-level state changes highlights its value as a continuous monitoring tool. It allows disaster management agencies to monitor exactly when these newly grown, low-risk fuel beds begin to cure and transition back into higher risk categories in subsequent years.

5.7. External Validation: Major Fire Footprints

To strengthen empirical credibility, the derived risk zones were overlaid with the footprints of major historical California fires (Fig. 7). The spatial intersection suggests a strong alignment between major burned areas and the “High” and “Extreme” risk clusters. To quantitatively support this visual assessment, a quantitative overlap metric (an Intersection over Union proxy) was calculated for the severe 2020–2021 mega-fire seasons. The results remarkably demonstrate that 99.81% of the total active fire points during these catastrophic events fell strictly within the boundaries of the ST-SCKM “Extreme” and “High” risk zones. This high quantitative overlap provides empirical support that the proposed framework captures key spatial patterns associated with conditions that facilitate large wildfire events [32].

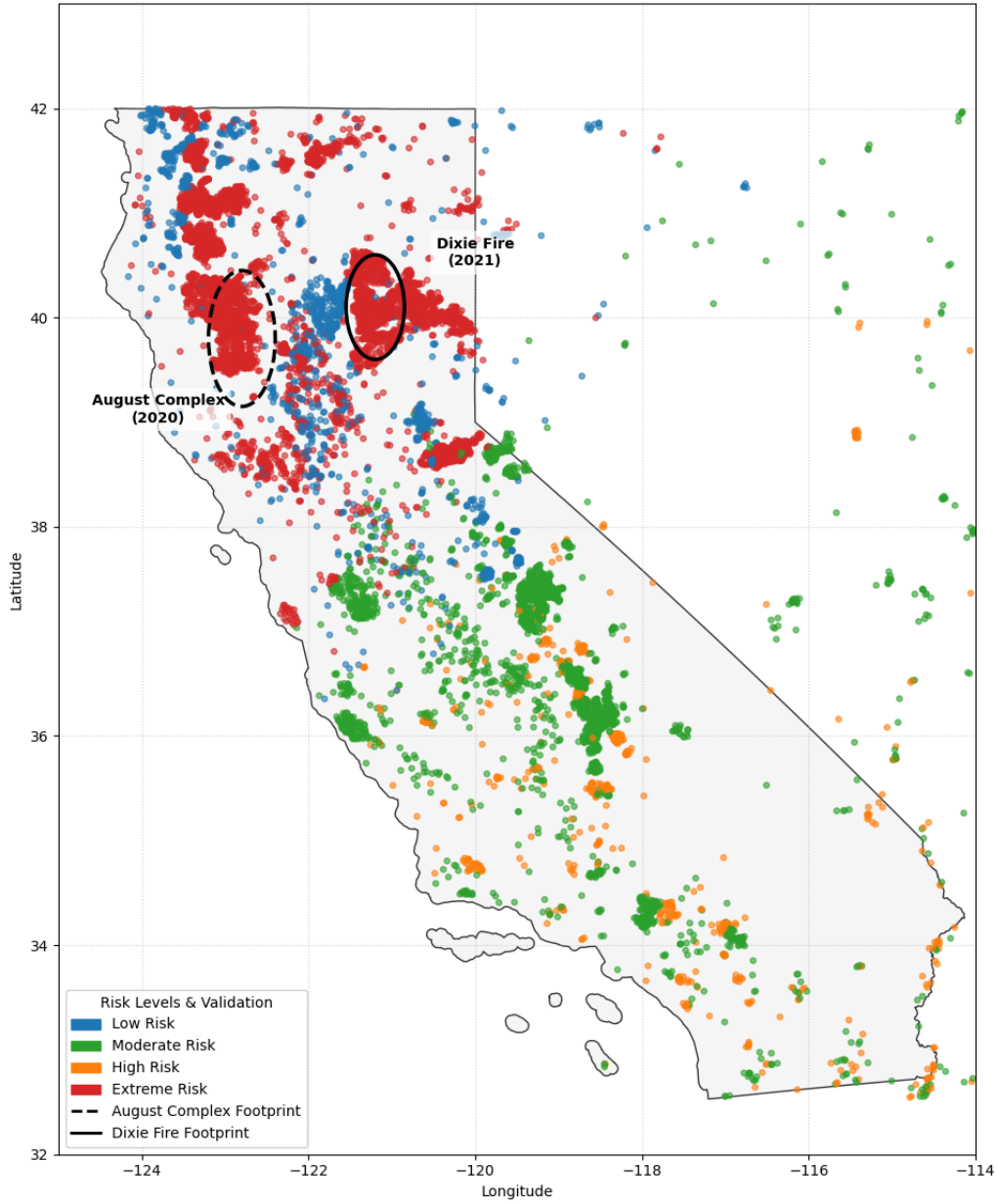


Fig. 7. External validation overlaying major historical California wildfire boundaries (2020–2021) onto the derived ST-SCKM risk zones, indicating alignment with major wildfire-affected areas.

The observed alignment highlights the effectiveness of the spatial contiguity penalty employed in the ST-SCKM optimization. Mega-fires, such as the Dixie and August Complex incidents [31], burned through dense, dry mixed-conifer forests under extreme meteorological stress. The fact that ST-SCKM classified these exact geographical coordinates as “Extreme Risk” prior to and during the peak ignition phases suggests that the framework may capture meaningful spatial patterns. This provides empirical evidence that the neighborhood-based penalty does not appear to impose arbitrary boundaries, but instead aligns with physically meaningful hazard zones.

Furthermore, this external validation has profound implications for Wildland-Urban Interface (WUI) management and evacuation planning. Because ST-SCKM generates cohesive, contiguous risk polygons, in stark contrast to the highly fragmented point-clouds generated by density-based methods like ST-DBSCAN, these outputs can be directly integrated with human settlement data. Emergency planners can overlay these validated

ST-SCKM risk boundaries onto census tracts, critical infrastructure maps, and road networks. This capability enables improved quantification of at-risk populations and facilitates the design of phased, safe evacuation routes, which are fundamental components of disaster resilience engineering.

5.8. Operational Implications for Resilience Planning

The link between this methodological framework and safety science is operational. Highly fragmented density-based clusters (ST-DBSCAN) are difficult for emergency agencies to utilize. By delineating contiguous, multi-dimensional risk zones, ST-SCKM provides actionable decision support.

To illustrate a concrete operational workflow, disaster management agencies such as CAL FIRE could directly ingest these contiguous zone polygons into their decision-support systems. During the pre-season (spring), forestry departments can overlay the persistent “High-Risk” polygons onto vegetation maps to systematically prioritize mechanical thinning and prescribed burn targets where fuel continuity is highest. Conversely, during late-summer red-flag warnings, the localized boundaries of the “Extreme-Risk” zones can be used to trigger automated, staged deployments of Very Large Air Tankers (VLATs) and to establish pre-calculated evacuation perimeters for adjacent Wildland-Urban Interface (WUI) communities, thereby transitioning from a reactive to a proactive resilience strategy.

5.9. Limitations and Future Work

While ST-SCKM provides a strong foundation for Geospatial AI-driven risk research, several limitations must be acknowledged. First, the fusion of 1 km fire point detections with 31 km daily ERA5 meteorological grids introduces scale mismatch and MAUP uncertainties. This inherently smooths out micro-topographical fire weather effects in complex terrains, introducing quantitative uncertainties and potential error propagation when localized ignition points are assigned coarse-grid meteorological values. Specifically, this scale mismatch may underrepresent highly localized drivers of extreme fire behavior, such as terrain-induced wind channeling in deep canyons or adiabatic heating along specific mountain ridges, meaning the meteorological profiles of our generated macro-zones represent broad synoptic trends rather than micro-scale physical conditions.

Second, the model currently relies entirely on active satellite fire detections. These satellite systems are susceptible to missing data and systematic detection biases, such as signal obscuration by dense cloud cover or thick smoke plumes generated by the fires themselves.

Finally, to address the operational forecasting utility of the framework, a forward validation test was conducted. The ST-SCKM model was trained exclusively on historical data (2019–2022) and subsequently used to predict the risk zones for the unseen future period (2023–2024). As illustrated in **Figure 8**, the evaluation of these forward projections revealed a logical, stepwise increase in hazard intensity. The areas classified into the “Extreme” and “High” danger zones exhibited mean Fire Radiative Powers (FRP) of 127.2 MW and 108.3 MW, respectively, drastically exceeding the baseline intensities of the predicted safe zones (Fig. 8a). Furthermore, the predictive feature space (Fig. 8c) confirms that these future Extreme-risk hotspots correctly align with severe meteorological drivers (high temperature and low humidity). This forward validation suggests that while ST-SCKM is an unsupervised algorithm, its spatio-temporal optimization inherently captures the physical precursors of severe wildfires, indicating its potential to generalize and identify high-risk areas in unseen future seasons.

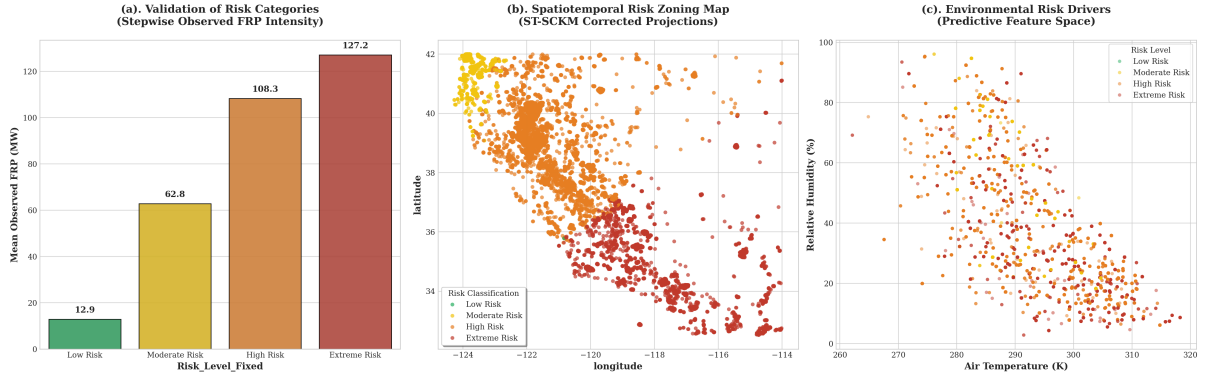


Fig. 8. Forward validation of the ST-SCKM framework (Predicting 2023–2024 based on 2019–2022 training data). (a) Stepwise validation confirming that predicted severe zones capture drastically higher empirical FRP intensities. (b) The spatial projection of the future risk zones across California. (c) The environmental predictive feature space demonstrating the alignment of Extreme-risk clusters with hot and dry meteorological conditions.

6. Conclusion

This study developed the Spatio-Temporal Spatially Constrained K-Means (ST-SCKM) framework to address the operational limitations of static, polygon-based regionalization and the spatial fragmentation inherent in density-based hotspot detection algorithms. By explicitly embedding a KNN-based spatial soft-penalty into the optimization objective, and by projecting continuous point data to a planar Cartesian system, ST-SCKM enables the delineation of coherent and contiguous wildfire risk zones.

Comprehensive hyperparameter tuning and evaluation on a massive 2019–2024 California dataset indicated that ST-SCKM shows improved performance under the evaluated parameter settings compared to both tuned ST-DBSCAN and Constrained Ward’s methods. It achieved a superior Silhouette Score (0.4728) and Calinski-Harabasz Index (7,113.98) while effectively incorporating 100% of the hazard observations without erroneously discarding critical detections as noise. The resulting contiguous macro-zones were logically profiled into discrete risk categories (Low to Extreme) driven by variations in temperature, humidity, and log-transformed Fire Radiative Power (FRP).

Critically, the framework’s utility for resilience engineering was validated through rigorous temporal and external analyses. Transition matrices quantified the strong spatio-temporal inertia of Extreme-risk zones during the severe drought years of 2020–2021, and they show strong spatial agreement with the empirical footprints of historic gigafires such as the August Complex and Dixie incidents. Conversely, the algorithm was able to capture the rapid contraction of severe risk zones during the 2023–2024 Atmospheric River events, suggesting sensitivity to macro-climatic conditions.

By decoupling widespread ignition potential from localized catastrophic intensity, and by generating cohesive polygons that can be integrated with Wildland-Urban Interface (WUI) data, the ST-SCKM framework provides a potential link between raw Geospatial AI hazard tracking and actionable emergency management. It can support disaster response agencies as a continuous monitoring tool for optimizing preseason fuel treatments, strategically pre-positioning aerial firefighting assets, and informing dynamic evacuation planning under increasingly volatile climate conditions.

Appendix A. Convergence Analysis of ST-SCKM

In this study, we propose the Spatio-Temporal Spatially Constrained K-Means (ST-SCKM) to ensure the formation of geographically contiguous risk zones. To validate the algorithm's reliability, we provide a theoretical analysis of its convergence properties. The ST-SCKM algorithm can be interpreted as a generalized K-means framework augmented with a graph-based spatial regularization term.

Definition A.1 (Objective Function). Let $X = \{x_1, \dots, x_N\}$ be the set of spatio-temporal observations. The ST-SCKM algorithm minimizes the objective function $J(\mathcal{C}, \mu)$, defined as:

$$J(\mathcal{C}, \mu) = \sum_{k=1}^K \sum_{i \in C_k} (w_s \|p_i - \mu_k^p\|^2 + w_t \|z_i - \mu_k^z\|^2) + \lambda \sum_{k=1}^K \sum_{i \in C_k} \sum_{j \in \mathcal{N}(i)} \mathbb{I}(l_i \neq l_j) \quad (\text{A.1})$$

where $\mathcal{C} = \{C_1, \dots, C_K\}$ denotes the cluster partition, $\mu = \{\mu_1, \dots, \mu_K\}$ denotes the set of centroids (with spatial and temporal components μ_k^p and μ_k^z), $\mathcal{N}(i)$ is the neighborhood set of observation i , and $\mathbb{I}(\cdot)$ is the indicator function. The parameters $w_s, w_t \geq 0$ are weights for spatial and temporal distances, and $\lambda \geq 0$ controls the strength of the spatial penalty.

Theorem A.1 (Boundedness). *The objective function $J(\mathcal{C}, \mu)$ is bounded below by 0. Proof.* The weighted squared Euclidean distance term $w_s \|p_i - \mu_k^p\|^2 + w_t \|z_i - \mu_k^z\|^2$ is non-negative. The spatial penalty term consists of a sum of indicator functions $\mathbb{I}(\cdot) \in \{0, 1\}$ multiplied by a non-negative parameter λ . Therefore, $J(\mathcal{C}, \mu) \geq 0$ for any valid partition $\mathcal{C}$ and centroid set μ . $\square$

Lemma A.1 (Monotonicity of the Centroid Update Step). *Given a fixed partition $\mathcal{C}$, updating the centroids μ does not increase J . Proof.* The spatial penalty term depends only on the cluster labels and is independent of the centroids. Therefore, minimizing J with respect to μ reduces to minimizing the weighted sum of squared errors. Since squared Euclidean distance is convex, the optimal centroid μ_k is given by the mean of the observations in cluster C_k . Thus, the update step ensures that $J(\mathcal{C}, \mu^{(t+1)}) \leq J(\mathcal{C}, \mu^{(t)})$. $\square$

Lemma A.2 (Monotonicity of the Assignment Step). *Given fixed centroids μ , updating the cluster labels $\mathcal{C}$ does not increase J . Proof.* In the assignment step, each observation x_i is assigned to the cluster that minimizes its total cost, defined as the sum of the spatio-temporal distance and the spatial penalty. Any reassignment that reduces this cost decreases the corresponding contribution to the objective function. Since each reassignment reduces the local contribution to J , the overall objective function is non-increasing, i.e., $J^{(t+1)} \leq J^{(t)}$. $\square$

Theorem A.2 (Convergence to a Local Minimum). *The ST-SCKM algorithm converges to a local minimum in a finite number of iterations. Proof.* From Lemma A.1 and Lemma A.2, the sequence of objective values $\{J^{(0)}, J^{(1)}, \dots\}$ is monotonically non-increasing. By Theorem A.1, this sequence is bounded below. Therefore, it converges to a limit value J^* . Since the number of possible partitions of the dataset is finite (K^N), and each iteration strictly decreases or preserves the objective value, the algorithm converges to a local minimum in a finite number of iterations. $\square$

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CRedit authorship contribution statement

Muh Akbar Idris: Conceptualization, Methodology, Software, Data curation, Formal analysis, Visualization, Writing - original draft. **Muhammad Nur Aidi:** Supervision, Validation, Writing - review & editing. **Anik Djuraidah:** Supervision, Validation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The datasets used in this study are derived from publicly available sources. Wild-fire event data were obtained from NASA's Fire Information for Resource Management System (FIRMS), available at <https://firms.modaps.eosdis.nasa.gov/>. Meteorological variables were obtained from the ERA5 reanalysis dataset provided by the Copernicus Climate Change Service (C3S) through the Climate Data Store, available at <https://cds.climate.copernicus.eu/>.

To facilitate transparency and reproducibility, the implementation of the proposed Spatio-Temporal Spatially Constrained K-Means (ST-SCKM) algorithm, together with example scripts and documentation, is publicly available on GitHub at:

<https://github.com/muhakbaridris/ST-SCKM>

Researchers are encouraged to use, reproduce, and extend the proposed framework in accordance with the repository license.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Google Gemini in order to improve language fluency, ensure academic formatting, and assist with LaTeX coding. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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